

Gyroscope

by

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What is Gyroscope?

Definition:

A gyroscope is a device for measuring or maintaining orientation, based on the principles of the conservation of angular momentum.

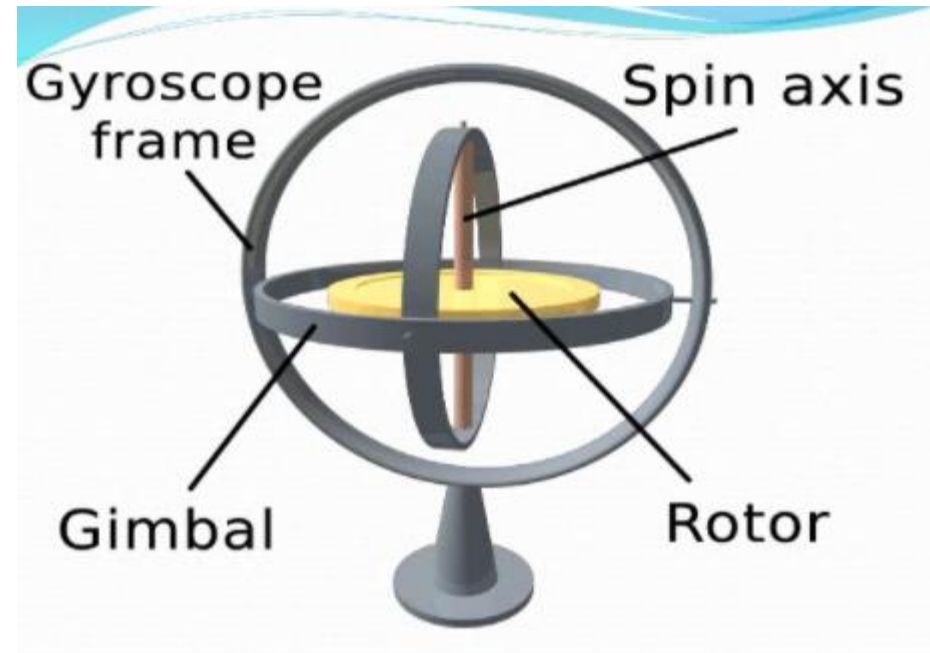


Fig. 1. Gyroscope

- If the axis of spinning or rotating body is given an angular motion about an axis perpendicular to the axis of spin, an angular acceleration acts on the body about the third perpendicular axis.
- The torque required to produce this acceleration is known as the active gyroscopic torque.
- A reactive gyroscopic torque or couple also acts similar to the concept of centripetal and centrifugal forces on a rotating body.
- The effect produced by the reactive gyroscopic couple is known as the gyroscopic effect.
- Thus aeroplane, ships, automobiles, etc, that have rotating parts in the form of wheels or rotors of engines experiences gyroscopic effect while taking turn, such as when axes of spin is subjected to some angular motions.

Gyroscope contd...

Precessional Motion:

- The forces acting on a spinning top.
- The torque produced is perpendicular to the angular momentum vector.
- This changes the direction of the angular momentum vector $\vec{L} \rightarrow$ according to $d\vec{L} = \vec{\tau} dt$, but not its magnitude.
- The top *precesses* around a vertical axis, since the torque is always horizontal and perpendicular to $\vec{L} \rightarrow$.
- If the top is *not* spinning, it acquires angular momentum in the direction of the torque, and it rotates around a horizontal axis, falling over just as we would expect.

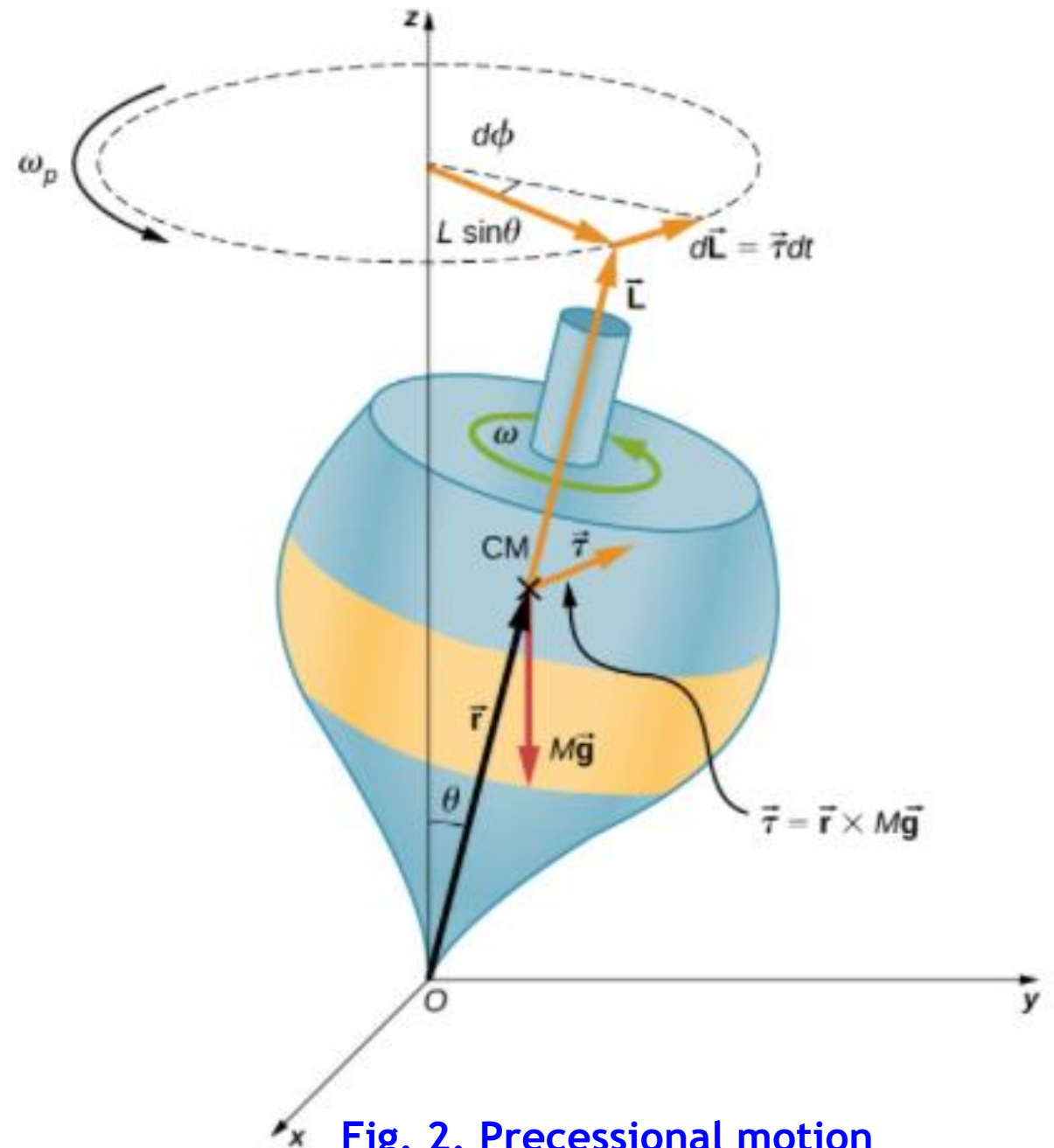


Fig. 2. Precessional motion

Gyroscope contd...

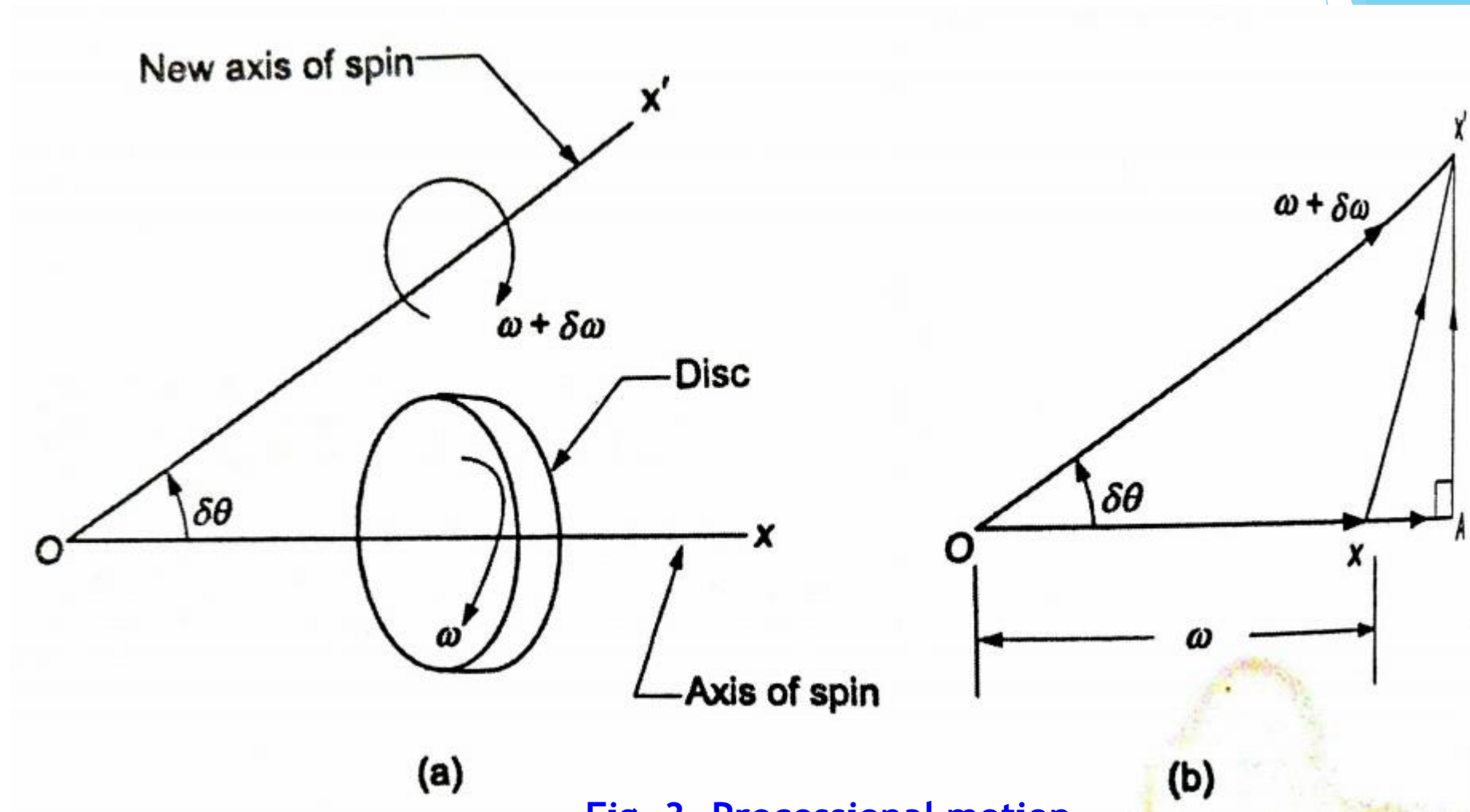


Fig. 3. Precessional motion

Gyroscope contd...

Angular acceleration along

$$\begin{aligned} OX &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega \cos \delta\theta + \delta\omega \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega + \delta\omega - \omega}{\delta t} \right] \quad (\text{Because } \cos \delta\theta \approx 1) \\ &= \lim_{\delta t \rightarrow 0} \frac{\delta\omega}{\delta t} \\ &= \frac{d\omega}{dt} \end{aligned}$$

Gyroscope contd...

Angular (or gyroscopic) acceleration perpendicular to OX ,

$$\begin{aligned} &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \sin \delta\theta}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \frac{(\omega \cdot \delta\theta + \delta\omega \cdot \delta\theta)}{\delta t} \\ &= \frac{\omega \cdot d\theta}{dt} \quad [\text{Neglecting } \delta\omega \cdot \delta\theta, \text{ being small}] \\ &= \omega \cdot \omega_p \end{aligned}$$

where ω_p is the precessional angular speed of the spin axis.
Total angular acceleration of disc = XX'



Gyroscope contd...

$$\begin{aligned} &= XA \quad AX' \\ &= \frac{d\omega}{dt} \quad \omega \cdot \frac{d\theta}{dt} \\ &= \frac{d\omega}{dt} \quad \omega \cdot \omega_p \end{aligned}$$

where $\rightarrow$ represents vector sum.

Gyroscope contd...

Total angular acceleration along XX' may be written as:

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

1. $d\omega/dt$, represents change in the magnitude of the angular velocity of the rotor.
2. $\omega.d\theta/dt$, represents change in the direction of the axis of spin, the direction of AX' is from A to X' in the vector diagram (being a component of XX'), the acceleration acts clockwise in the vertical plane xz (when viewed from front along y -axis).

Gyroscope contd...

Fundamentals of Gyroscopic Motion:

Gyroscope: It is a body which spinning about an axis is free to move in other direction under the action of external forces.

Axis of Spin: It is axis about which the body revolves.

Gyroscopic effect: consider a body spinning about an axis OX . If couple represented by vector OZ perpendicular to OX .

- Then the body tries to precess about an axis OY , which is perpendicular to both OX and OZ .
- This combined effect is called gyroscopic or precessional effect.
- The plane of spin, plane of precession, and plane of gyroscopic couple are mutually perpendicular.

Gyroscope contd...

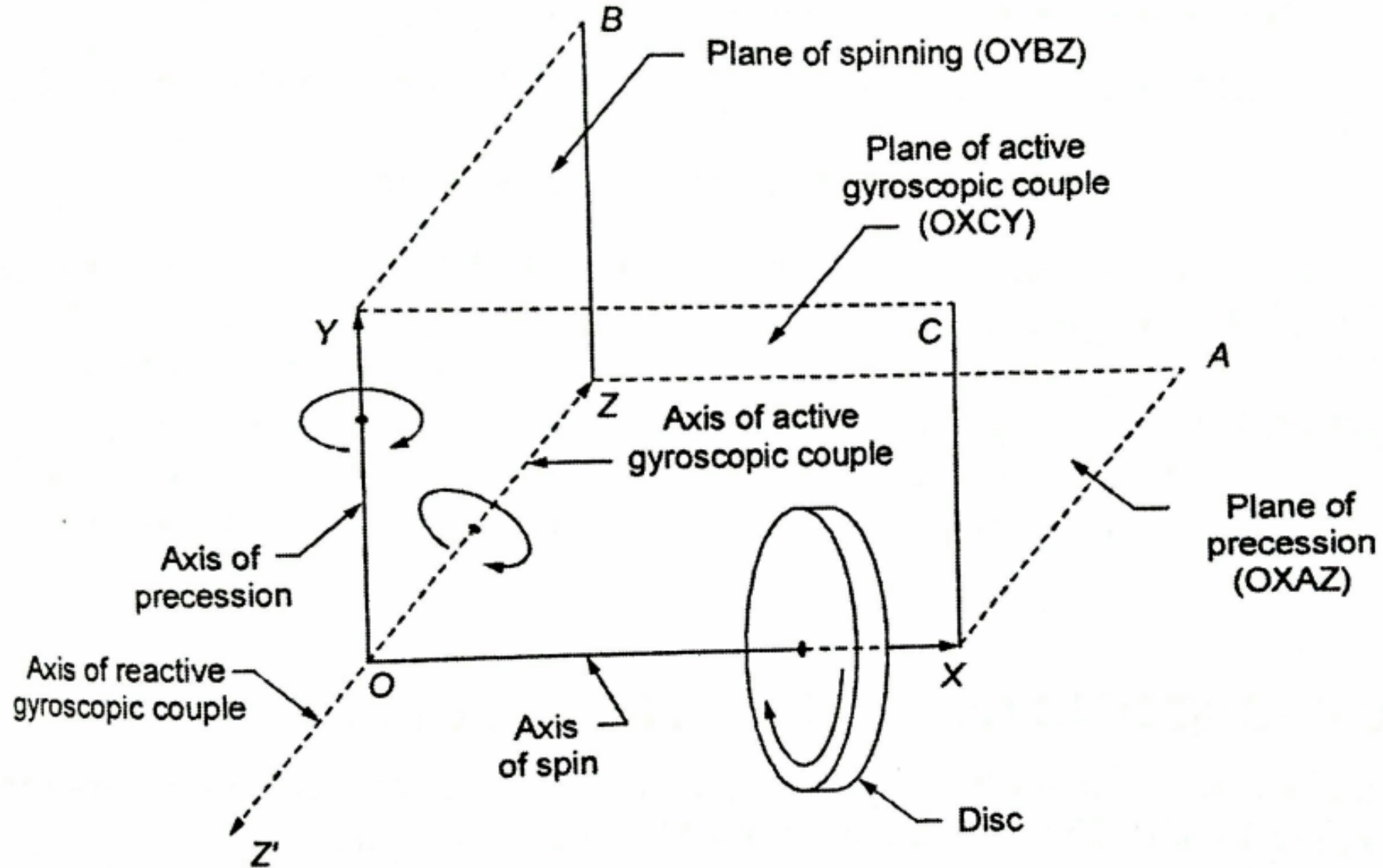


Fig. 4. Gyroscopic effect

Gyroscope contd...

Fundamentals of Gyroscopic Motion:

Precession: It means the rotation about the third axis OY , which is perpendicular to both the spin axis OX and the couple axis OZ .

Axis of Precession: The third axis OY about which the body revolves and is perpendicular to both the spin axis OX and couple axis OZ , is called the axis of precession.

Mechanical Gyroscope: It is a special mechanism generally employed for the control of angular motion of a body.

- Gyroscope is an instrument, which appears as a device processing intelligence.
- Gyroscopic effects can be used in several applications for directional control, i.e. in Gyrocompass used on aeroplanes and ships and in inertia guidance control system for missiles and space craft control.

Gyroscope contd...

Gyroscopic Couple:

Torque or Couple C:

$$T = I \frac{d\omega}{dt} + T\omega \frac{d\theta}{dt}$$

$$T = I\alpha + T\omega\omega_p$$

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

Gyroscopic Couple

Example 17.1 A uniform disc having a mass of 8 kg and a radius of gyration of 150 mm is mounted on one end of a horizontal arm of 200-mm length. The other end of the arm can rotate freely in a universal bearing. The disc is given a clockwise spin of 250 rpm as seen from the disc end of the arm. Determine the motion of the disc if the arm remains horizontal.

Solution $m = 8 \text{ kg}$ $l = 0.2 \text{ m}$ $k = 0.15 \text{ m}$ $N = 240 \text{ rpm}$ $I = mk^2 = 8 \times (0.15)^2 = 0.18 \text{ kg} \cdot \text{m}^2$

$$\omega = \frac{2\pi \times 240}{60} = 25.13 \text{ rad/s}$$

$$C = I \omega \omega_p$$

$$Mgl = I \omega \omega_p$$

$$8 \times 9.81 \times 0.2 = 0.18 \times 25.13 \times \omega_p$$

$$\omega_p = 3.47 \text{ rad/s}$$

As the disc rotates, the weight of the disc acts downwards and thus a couple about the y-axis, in the clockwise direction is applied on the disc (Fig. 17.4). The reaction couple is thus counterclockwise which tends to keep the arm horizontal.

Assuming that the axis of spin Ox precesses about the z-axis in the counterclockwise direction, the vector oa would rotate to the position ob in a short period. Then ab is the applied couple and $b'a'$ is the reaction couple which is clockwise and tends to give the arm a clockwise rotation about the y-axis which is not true. Thus, the axis of spin Ox must precess about the z-axis in the clockwise direction with an angular velocity of 3.47 rad/s.

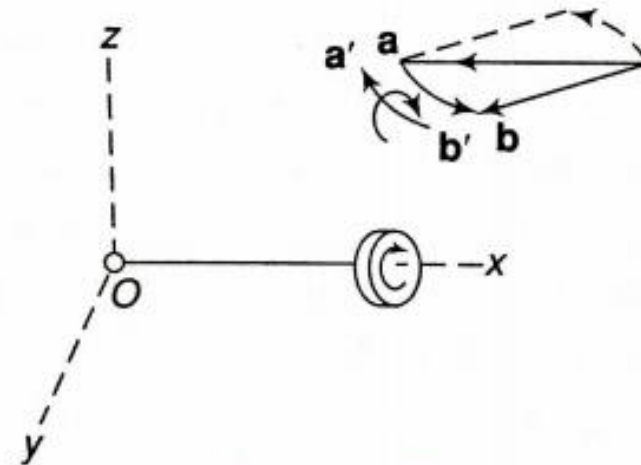


Fig. 17.4

Example 17.2 A disc with radius of gyration of 60 mm and a mass of 4 kg is mounted centrally on a horizontal axle of 80 mm length between the bearings. It spins about the axle at 800 rpm counterclockwise when viewed from the right-hand side bearing. The axle precesses about a vertical axis at 50 rpm in the clockwise direction when viewed from above. Determine the resultant reaction at each bearing due to the mass and the gyroscopic effect.

Solution $m = 4 \text{ kg}$ $N = 800 \text{ rpm}$ $k = 0.06 \text{ m}$ $N_p = 50 \text{ rpm}$ $I = mk^2 = 4 \times (0.06)^2 = 0.0144 \text{ kg.m}^2$
 $l = 80 \text{ mm} = 0.08 \text{ m}$

$$\omega = \frac{2\pi \times 800}{60} = 83.78 \text{ rad/s}$$

$$\omega_p = \frac{2\pi \times 50}{60} = 5.24 \text{ rad/s}$$

$$\therefore C = I \omega \omega_p = 0.0144 \times 83.78 \times 5.24 = 6.32 \text{ N.m}$$

The applied (active) and reaction couples are shown in Fig. 17.5. The reaction couple is clockwise when viewed from front and tends to raise the bearing A and lower the bearing B . Thus, reaction of each bearing in turn is downwards at A and upwards at B .

Reaction at bearing A due to gyro. couple

$$= \frac{C}{l} = \frac{6.32}{0.08} = 79 \text{ N (downwards)}$$

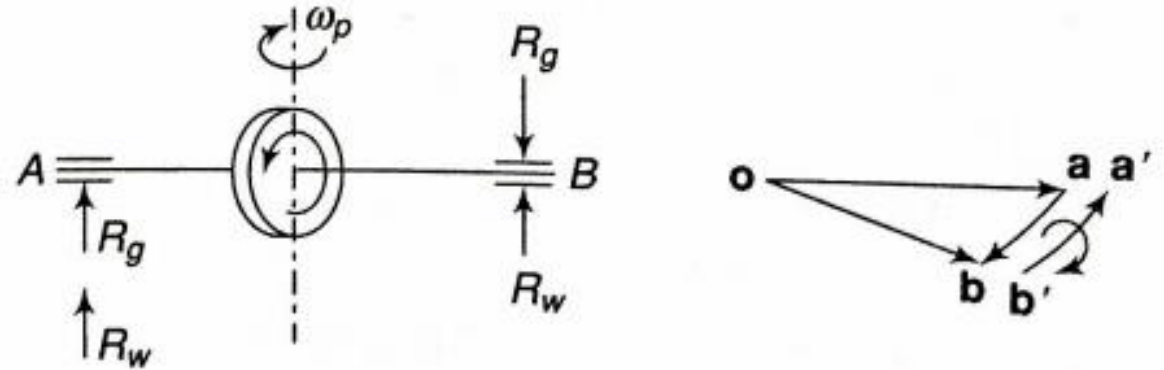


Fig. 17.5

Force at bearing B due to gyro. couple = 79 N (downwards)
(Reaction at each bearing is to be in the opposite direction.)

$$\text{Force at each bearing due to weight of the disc} = \frac{4 \times 9.81}{2} = 19.6 \text{ N}$$

or Reaction at each bearing due to weight = 19.6 N (upwards)

$$\therefore \text{Reaction at bearing } A = 79 - 19.6 = 59.4 \text{ N (upwards)}$$

$$\text{Reaction at bearing } B = 79 + 19.6 = 98.6 \text{ N (downwards)}$$

Thank You

